

EMBRY-RIDDLE Aeronautical University

Computer Architecture DB-CESC 470

Section(s): 01DB

Course Information

Academic Term: Daytona Beach Fall 2026

Term Dates: August 24, 2026 - December 9, 2026

Credit Hours: 3

Mode of Delivery: In Person

Meeting(s): 11:00 AM - 11:50 AM Monday, Wednesday, Friday

Location(s): Lehman Building 369

Instructor Information

Name

Siyao Li

Email

lis14@erau.edu

Catalog Course Description

Detailed exploration of Von Neuman computer architecture, including processors, memory, input/output, machine language, assembly language, microprogramming, and operating systems. Coverage of advanced topics in computer architecture and systems.

Course Goals

Students will understand the trade-offs behind modern computer architecture design, learning key concepts like instruction sets, memory hierarchy, and pipelining.

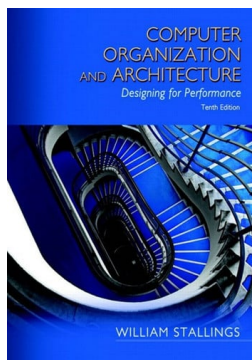
Student Learning Outcomes

1. Evaluate the principles and trade-offs (cost/performance, speed/flexibility) behind the design of modern computer systems, using both qualitative and quantitative methods.
2. Design an instruction set for a specific processor architecture, considering factors such as complexity, performance, and flexibility.
3. Analyze hierarchical memory system designs, including cache performance and optimization strategies, to improve system efficiency.
4. Optimize processor performance using applicable concepts of basic and advanced pipelining techniques.
5. Assess the advantages and limitations of hardwired and microprogrammed control mechanisms in modern computer architectures.
6. Explain how the architecture of multicore processors improves parallelism and system throughput.

Course Prerequisites

CESC 470 Prerequisite: CESC 320 and CESC 320L (CE/SE students must have C or better).

Required Course Materials



Text/Tool: Computer Organization and Architecture

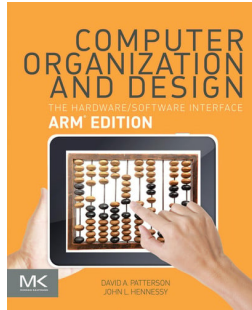
ISBN: 9780134102061

Authors: William Stallings

Publisher: Pearson Education

Publication Date: 2015-01-16

Edition: 10



**Text/Tool: Computer Organization and Design
ARM Edition**

ISBN: 9780128018354

Authors: David A. Patterson; John L. Hennessy

Publisher: Elsevier S & T

Publication Date: 2016-05-06

Edition: 1

Grading

Letter Grade	Percentage
A	90 - 100%
B	80 - 89%
C	70 - 79%
D	60 - 69%
F	< 60%

University Policies

ACADEMIC INTEGRITY

Embry-Riddle is committed to maintaining and upholding academic integrity. Academic integrity violations include cheating, fraud, plagiarism, and double-submissions.

HONOR CODE

Per the terms of the Embry Riddle Aeronautical University Student's [Honor Code](#), the following behavior is unacceptable and will not be tolerated and may result in disciplinary action as defined by University Policy.

- Abusive, Threatening or Bullying Behavior
- Discrimination
- Harassment

Class Attendance

Class attendance is an important component of the educational process and is directly linked to student success. Students who are unable to

attend class due to illness, injury, bereavement, accident, or other extenuating circumstances must notify their instructor as soon as possible.

The University's [Absence Verification](#) Process allows students to request official verification of qualifying absences through the Dean of Students Office. Absence verification serves as confirmation of the reported circumstance but does not, in itself, excuse the absence. Excused absences are determined solely by the instructor in accordance with the class attendance policy. Students are responsible for making up any missed work as permitted by the instructor.

Form link:

[Daytona Beach - Verification of Absence](#)

Leave of Absence

The University's Leave of Absence (LOA) Policy (voluntary or involuntary) allows students to temporarily pause their enrollment to address personal, medical, financial, military, or other significant matters without jeopardizing their future standing at the University. The policy is designed to support student-wellbeing, uphold academic integrity, and maintain a safe, supportive educational environment for all. A Leave of Absence must be approved by the Dean of Students Office and may be granted for a period of one to four academic semesters (Fall and Spring terms only).

Use of Artificial Intelligence

The use of ChatGPT and other artificial intelligence (AI) tools in an educational setting must be guided by ethical principles and academic integrity. Students and faculty may use these tools for research support or aids to enhance their learning and scholarship but should not rely on them to produce original work. The University recognizes the potential benefits of AI but acknowledges the potential risks, such as creating unauthentic, inaccurate, biased, or harmful content. Students should follow best practices to ensure authentic scholastic accomplishment and academic integrity, including **avoiding plagiarism or machine ghostwriting**.

Ultimately, the University upholds the importance of intellectual honesty and ethical research practices and expects students to act in accordance with these principles when using AI tools.

Weather Emergency Statement

Hurricanes, tornadoes and other natural disasters (such as fires) are a part of life in Florida. In the event a natural disaster threatens our area, everyone at ERAU is expected to monitor voicemail, email, and the local media for any changes to the normal schedule, including evacuation plans. Decisions to close the Daytona Beach campus are typically made sometime in the afternoon on the day before the intended closure. In the event of an emergency during class hours, please listen carefully to directions from your instructor. If it becomes necessary to evacuate the classroom, we will gather at a designated meeting point away from the building and take attendance to ensure everyone is safe and accounted for. As part of the disaster preparedness process, it is strongly suggested that each member of the ERAU community enroll in the RAVE emergency notification system. If you have not done so, please sign up using the link provided on your ERNIE home page.

Course Policies

Office Hours: Instructors are available during scheduled office hours for questions and additional support. Appointments outside office hours can be arranged by request.

Homework Submission: All homework must be typed and converted to pdf for submission.

Attendance: Attendance is not recorded, but it is factored into the quizzes for the class. It should be noted that a strong correlation exists between being attentive in class and performance on course examinations.

Late Penalty: Assignments submitted after the deadline will incur a penalty of 20% within 3 days. After 3 days, late submissions will not be accepted unless prior arrangements have been made.

Extensions: Extensions may be granted in exceptional circumstances (e.g., illness, family emergency). Requests must be made before the deadline.

Accessibility: Students with disabilities are entitled to reasonable accommodations. Please inform the instructor at the beginning of the course, and provide the necessary documentation to arrange accommodations.

Important Dates: A tentative schedule for homework assignments and quizzes will be provided later. Please review it and inform me of any time conflicts.

Make-Up Exams Eligibility: Make-up exams are allowed only for students who provide valid documentation for their absence (e.g., medical reasons).

Make-Up Exams Scheduling: Make-up exams must be scheduled within 7 days of the original exam date.

Plagiarism: Plagiarism, cheating, and other forms of academic dishonesty are strictly prohibited. Any violation will result in disciplinary action, which may include a failing grade for the assignment or course.

Collaboration: While collaboration on certain projects may be encouraged, all submitted work must be your own unless otherwise stated. Always cite sources appropriately.

Student Resources

CAMPUS SAFETY & SECURITY

[Campus Safety & Security](#) officers are on duty 24 hours per day, 365 days a year. We strongly encourage students to report crimes, emergencies, or suspicious conditions to Safety & Security by calling the department's

Communications Center at **386-226-6480**. In the event of an emergency, call **386-226-SAFE (7233)**.

CANVAS HELP

When logged in to Canvas, click Help – located at the bottom of the global navigation on the left side. This menu has choices for Canvas Support that you can choose from:

- Search the Canvas Guides
- Canvas Support Hotline at **1-833-334-2831**, available 24 hours a day, 7 days a week
- [Chat with Canvas Support](#)

COUNSELING CENTER

The [Counseling Center](#) is located in the Wellness Center Complex, building 502

Phone: **386-226-6035** Monday - Friday, 8am - 5pm

Social media accounts (@eraucounseling)

24/7 Mental health crisis support via app, telephone or web is also available through Telus Health Student Support, which is available through App Store/Google Play.

STUDENT ACCESSIBILITY SERVICES

Embry-Riddle provides reasonable accommodations for students with documented disabilities through Student Accessibility Services located in Building 500. To request accommodations, or ask any additional questions, please call **386-226-7916** or email dbsas@erau.edu.

[Student Accessibility Services Office](#)

THE CENTER FOR FAITH AND SPIRITUALITY/CHAPLAINS OFFICE

The [Center for Faith and Spirituality](#) is open from 6 a.m. until midnight; Reverend David Keck and Father Karl Bergin work with students of all faiths as well as those from no faith tradition, to offer confidential, individual meetings to discuss life's challenges; offer spaces for quiet reflection; and opportunities to learn about other traditions.

For more information or to schedule a meeting: chaplain@erau.edu.

FOOD PANTRY

We believe access to food is a human right, accessible to every student. We are committed to educating and distributing aid to our Embry-Riddle community. The mission of the Food Pantry program is to support the academic success of students by providing supplemental food for those who are experiencing food insecurity. Students can apply for support through the [Food Pantry Request form](#). A time will be scheduled with the student once the request is received.

HEALTH SERVICES

[Daytona Beach Health Services](#) clinic is located in Building 500 on the corner of Richard Petty and Clyde Morris Blvd. Our team of medical professionals, including a University Physician, Licensed Dietitian, Nurses and Insurance Specialist, are dedicated to supporting your health and well-being. Whether you need medical care, have questions or require assistance, please do not hesitate to contact us at **386-226-7917** or dbhealth@erau.edu. We offer our services on a walk-in or appointment basis, and we are here to support your health needs.

HUNT LIBRARY

The Hunt Library is here to help you succeed with finding just the right information resources. For detailed research assistance, please contact [Ask a Librarian](#).

- Website: <https://hunt-answers.erau.edu/ask>
- Email: library@erau.edu
- Phone: **386-226-7656 | 800-678-9428**

INTERNATIONAL STUDENT & SCHOLAR SERVICES

International students with questions about I-20s, visas or other related services that assist with the maintenance of their status and immigration compliance should contact [International Student & Scholar Services](#) at **386-226-6579** or dbiss@erau.edu. For other international student related issues and questions such as health insurance, taxes, drivers' licenses, campus work authorization, please contact the International Programs Administration Office at **386-323-8133**. Both offices are located on the first floor of New Residence Hall 3.

STUDENT ATHLETE SERVICES

[Faculty Athletic Representative \(FAR\)](#)

Dr. Jennifer Hinebaugh

Email: hinebb07@erau.edu

Phone: 386-226-6564

STUDENT GOVERNMENT

[Student Government Association \(SGA\)](#) proudly offers a variety of services to improve student life on campus. Through its three branches, SGA can fund student organizations, advocate for students, and carry out multiple projects. Besides giving students a voice on campus, the SGA also has four divisions including: *The Avion*, Touch-N-Go Productions, Emergency Response Team, and WIKD radio station.

TITLE IX

Title IX prohibits sex discrimination, sexual harassment, and sexual misconduct. Anyone may report suspected violations to the Title IX Office.

Please visit the Title IX website for additional information.

Website: dbtix.erau.edu

Email: dbtitle9@erau.edu

Office: Student Union, Suite 324

[Online Complaint Form](#)

UNDERGRADUATE RESEARCH

Embry-Riddle strives to create a culture of knowledge discovery through research. The [Office of Undergraduate Research](#) engages undergraduate students in faculty-mentored research that is both faculty and student-led. Our mission is to provide a diverse set of opportunities for all undergraduate students to enhance their education through engagement in research, inquiry, innovation, and/or other scholarly projects.

VETERANS STUDENT SERVICES

Being a student veteran can result in a variety of complexities that might require accommodations. Complications with VA benefits disbursements, and other unforeseen military-related developments can complicate your academic life. Therefore, please consider making professors aware of your Veteran status and contacting Veterans Student Services.

[Veteran Student Services \(VSS\)](#) facilitates the transition of military-affiliated students from military culture to University life, supports their academic success through informative programming, and assists veterans, active service members, guardsmen, reservists, and military dependents in receiving their military educational benefits. The unit is

staffed with qualified school certifying officials who deliver a broad range of services; while providing informal counseling to students using VA educational benefits. In addition, VSS works in collaboration with our [Student Veterans of America chapter](#) and [Faculty-4-Veterans](#), who support the unit in addressing the needs of our military-affiliated students across campus. VSS, which is located in Residence Hall 3, Bldg. 255, can be reached at **386-226-6350** or dbva@erau.edu.

Tutoring Services

COLLEGE OF ARTS AND SCIENCES TUTORING

Academic Advancement Center

The [Academic Advancement Center](#) (AAC) is your direct connection to academic support, which provides free tutoring and Supplemental Instruction (SI) to all DB students on a walk-in basis (no appointments needed). The AAC provides peer led tutoring for all foundational math, physics, engineering, chemistry, and biology courses. The AAC is located on the first floor of New Residence Hall 3 (NRH3).

Please refer to the [Academic Advancement Center website](#) for details and schedules (<https://campusgroups.erau.edu/erauaac/home/>) and consider following the AAC's Instagram account for updates @erauaac.

Fall & Spring Hours of Operation:

- Monday – Thursday: 9:00 AM – 9:00 PM
- Friday: 9:00 AM – 5:00 PM
- Sunday: 4:00 PM – 8:00 PM

Math Tutoring Lab located in NRH3-112 (MATH 111, MATH 112, MATH 143, MATH 241, MATH 242, and MATH 243)

Physics Tutoring Lab located in NRH3-118 (PHYS 113, PHYS 117, PHYS 150, PHYS 160, PHYS 250, PHYS 226, PHYS 227, and PHYS 228)

Engineering Tutoring Lab located in NRH3-113 (ENGR 115, ENGR 120, ENGR 201, ESCI 202, ESCI 204 Dynamics, and AERO 201)

Chemistry Tutoring Lab located in NRH3-126 (CHEM 110 and CHEM 110L)

Biology Tutoring Lab located in COAS 417 (BIOL 120, BIOL 305, BIOL 306, BIOL 321, BIOL 322, and BIOL 405)

General Study Area located in NRH3-119.

This area is available for all students to use as a general study space. Utilizing this room encourages students to collaborate with their peers and form study groups at their convenience.

Computer Lab located in NRH3-117.

This lab is open to all students and is equipped with standard academic university software and free printing.

MAP LAB located in COAS 301.28

The Math Department provides support for ALEKS and MyLab Math Software and tutoring. (MATH 06, MATH 143, MATH 241, MATH 242, MATH 243, and upper-level math courses.)

Center for Communication and Digital Media

The Center for Communication and Digital Media (CCDM) combines the former Writing Center and the former Digital Studio to offer comprehensive communication tutoring services. The CCDM offers peer tutoring that adapts to the diverse needs of every communicator. We support students with written, spoken, and digital communication, and the CCDM offers access to the Adobe Creative Cloud, Final Cut Pro, professional audio-visual recording and editing equipment, and a variety of other software and hardware. Appointments can be made

here: <https://erau.mywconline.com>. You must create an account in order to schedule an appointment. The CCDM is located in SU 431, inside the Hunt Library.

COLLEGE OF AVIATION TUTORING

The **Aviation Learning Center (ALC)** located in COA 141 provides free tutoring on Aeronautical Science and Flight course related topics.

Fall/Spring Hours of Operations: Monday - Friday, 8 a.m. - 8 p.m.;

Saturday 10 a.m. - 5 p.m. Summer hours: 10 a.m. - 5 p.m

The ALC offers access to ten Advanced Aviation Training Devices (AATDs) that are able to simulate both the Cessna 172S and the DA-42 VI, including the Garmin G-1000 avionics, and is equipped with training aids including aircraft components, cut-away instruments, cockpit procedure trainers, Flight Management Systems trainers, reference literature, and much more.

Weather Tutoring Lab is open Monday through Friday from 9:00 AM – 5:00 PM and staffed with both virtual and face-to-face tutors who can help you with many meteorology, math and physics courses. The weather lab is located in COA 356.

COLLEGE OF ENGINEERING TUTORING

Provides tutoring for upper-level 300 & 400 level engineering courses (ENGR, ESCI, AERO, CIVE, CPSS, CESC, ELEC, SFTW, SYSE, and MECH). The tutoring lab is located in COE LB 332.

COLLEGE OF BUSINESS TUTORING

COB offers tutoring in Accounting, Management, Economics, Business Writing, Product Management, Finance, Marketing, and Business Analytics. Tutoring is located in COB 271 from Monday to Thursday from 10:00 AM – 5:00 PM in the following courses.